ZORIAN THORNTON

Ph.D, Seattle, WA

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RESEARCH INTERESTS

I develop machine learning models for problems in molecular biology and immunology. My interests span the adaptive immune system and viral antigenic evolution, centered on the antibodies and antigens that mediate their interaction. I am particularly focused on applying protein language models and computational protein design methods to understand and engineer these molecules.

EDUCATION

University of Washington - Seattle, WA *Sep. 2019 - March 2026*
Ph.D in Genome Sciences
Dissertation: *Computational tools for studying viral antigenic evolution*

Virginia Tech - Blacksburg, VA *Aug. 2015 - May 2019*
B.Sc Statistics

Virginia Tech - Blacksburg, VA *Aug. 2015 - May 2019*
B.Sc Computational Modeling and Data Analytics

Minors in Mathematics and Computer Science

TECHNICAL SKILLS

Languages	Python, R, Java, SQL, MATLAB, bash
ML & Statistics	PyTorch, TensorFlow, JAX, NumPyro, scikit-learn; supervised and unsupervised learning, deep learning, Bayesian inference, probabilistic programming, MCMC, variational inference, hierarchical modeling, dimensionality reduction, model benchmarking
Computational Biology	Phylogenetic inference, sequence analysis, deep mutational scanning analysis, antigenic evolution modeling; Augur, Auspice, Biopython, evofr
Scientific Computing	pandas, numpy, scipy, matplotlib, seaborn; SLURM, CUDA, openMP, AWS, Redshift, Unix/Linux, git, high-performance computing

RESEARCH AND WORK EXPERIENCE

Genentech *South San Francisco, CA*
Data Science Intern, Evidence for Access (Technical Institute) *June 2026 - September 2026*

- Implemented and deployed a machine learning model (CatBoost) for imputation of proxied costs on 100M+ claims records, supporting Health Economics Outcomes Research (HEOR) and Real-World Data (RWD) analyses

Fred Hutchinson Cancer Research Center *Seattle, WA*
Doctoral Researcher, advised by Frederick Matsen IV *July 2021 - June 2026*

- Developed open-source simulation framework for modeling viral genetic and antigenic evolution under immune selection pressure
- Designed biophysically-inspired neural network architectures for predicting protein variant phenotypes from deep mutational scanning data
- Applied Bayesian statistical methods and probabilistic programming to benchmark viral frequency forecasting models

Adaptive Biotechnologies *Seattle, WA*
Machine Learning Computational Biology Intern *June 2022 - Sept. 2022*

- Implemented and benchmarked semi-supervised learning methods for classification of T-cell receptor specificity from high-throughput sequencing data, characterizing model performance across data conditions through systematic downsampling experiments
- Built data preprocessing pipelines for large, complex biological sequence datasets and visualization tools to communicate findings to experimental and computational collaborators

Virginia Tech Department of Statistics *Blacksburg, VA*
Undergraduate Researcher, advised by Leah Johnson *Aug. 2017 - May 2019*

- Built Bayesian hierarchical models using MCMC methods to quantify uncertainty in environmental-disease relationships, incorporating temperature-sensitive vector life history traits and host-pathogen interactions

Nielsen *Chicago, IL*
Data Science Intern *June 2018 - Aug. 2018*

- Implemented statistical framework to identify possible errors in scanned receipt data and implemented pipeline to attempt to correct errors

University of Washington Department of Genome Sciences *Seattle, WA*
Undergraduate Researcher, advised by Jim Bruce *June 2017 - Aug. 2017*

- Built probabilistic models to quantify competition between proteins for cross-linking, informing characterization of protein function and protein-protein interactions
- Developed 3D visualization tools for inter-surface regions of cross-linked proteins, contributing to published proteomics software

PUBLICATIONS

- **Thornton, Z.**, Tran, T., Figgins, M., Huddleston, J., Bedford, T., Matsen IV, F.A., Haddox, H. (2026). Simulating the genetic and antigenic evolution of viruses under selection pressure from host immunity. *Manuscript under revision at Virus Evolution.*
- Yu, T.*, **Thornton, Z.***, Hannon, W., DeWitt, W.S., Radford, C., Matsen IV, F.A. and Bloom, J.D. (2022). A biophyscial model of viral escape from polyclonal antibodies. *Virus Evolution*, 8(2), veac110.
- El Moustaid, F., **Thornton, Z.**, Slamani, H., Ryan, S. and Johnson, LR (2021). Predicting temperature-dependent transmission suitability of bluetongue virus in livestock. *Parasites Vectors*, 14(1), pp.1-14..
- Keller, A., Chavez, J.D., Eng, J.K., **Thornton, Z.** and Bruce, J.E. (2018). Tools for 3D Interactome Visualization. *Journal of proteome research*, 18(2), pp.753-758.

*:These authors contributed equally to this work.

RESEARCH PRESENTATIONS

- Zorian Thornton, *Predicting Disease Progression of Colorectal Cancer via Self-Organizing Maps*, RECOMB 2019, George Washington University, Washington DC, May 2019
- Zorian Thornton, *Modeling Bluetongue Virus via Markov Chain Monte Carlo Methods*, Student Experiential Learning Conference, Virginia Tech, Blacksburg, VA, Apr. 2018

- Zorian Thornton, *Viewing Molecular Interaction Interfaces Through Directed Computational Methods*, Department of Genome Sciences Research Symposium, University of Washington, Seattle, WA, Aug. 2017

TEACHING AND MENTORING

Graduate Teaching Assistant, University of Washington *2022 - 2023*

- Genome 361: Fundamentals of Genetics and Genomics (Spring 2022)
- Genome 475: Debates in Genetics (Spring 2023)

Research Mentor, University of Washington *2022 - 2024*

- Mentored Thien Tran (B.S., Paul G. Allen School of Computer Science), now a software engineer at Amazon

HONORS AND AWARDS

National Science Foundation Graduate Research Fellowship Program, Honorable Mention *2020*
Mu Sigma Rho, National Honor Society for Statistics *2019*
College of Science Dean's Roundtable Scholarship, Virginia Tech *2018–2019*
Luther and Alice Hamlett Undergraduate Research Grant, Virginia Tech *2018–2019*
Fralin Undergraduate Research Fellowship, Virginia Tech *2017–2018*
Eckert Statistics Scholar, Virginia Tech *2016–2019*